

Corey T. White, Ph.D.

Geospatial Research Software Engineer, Center for Geospatial Analytics, North Carolina State University
Founder and CEO, OpenPlains Inc.

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PROFESSIONAL SUMMARY

Geospatial research software engineer and core contributor to the open-source GRASS project. Work spans participatory geospatial modeling, geomorphometry, hydrological simulation, post-disaster topographic change detection, and interactive 3D visualization in the cloud. First author of the forthcoming "Geomorphometry in GRASS" chapter in the second edition of Reuter, Grohmann, and Lecours's *Geomorphometry: Concepts, Software, Applications* (Elsevier, 2026). Founder of OpenPlains Inc., an open-source platform for scenario-based geospatial modeling and simulation for community resilience planning.

EDUCATION

Ph.D., Geospatial Analytics May 2023

North Carolina State University, Raleigh, NC

- Dissertation: *Facilitating Civic Engagement in Rapidly Urbanizing Regions through Geospatial Participatory Modeling*
- Co-advisors: Dr. Helena Mitasova and Dr. Ross K. Meentemeyer
- ProQuest ISBN: 9798379880231

M.S., Geographic Information Systems and Technology May 2015

North Carolina State University, Raleigh, NC

B.A., Anthropology and Sociology May 2010

Warren Wilson College, Asheville, NC

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Geospatial Research Software Engineer Aug 2023 to present

Center for Geospatial Analytics, North Carolina State University

- Senior Personnel, NSF POSE Phase II: Growing GRASS OSE for Worldwide Access to Multidisciplinary Geospatial Analytics
- Lead researcher on USDA NRCS project incorporating USDA soils data into SIMWE (SIMulation of Water Erosion), a process-based overland flow and erosion model
- Co-led collaboration with the U.S. Army Corps of Engineers to rapidly generate digital surface models for Hurricane Helene disaster response
- Instructor of record for GIS 582 Geospatial Modeling and Analysis and for Mapping and Analytics with UAS
- Led National I-Corps cohort conducting 70+ stakeholder interviews to guide the strategic direction of GRASS

Founder and CEO Mar 2023 to present

OpenPlains Inc., Durham, NC

- Launched OpenPlains Inc. to deliver an online open-source geospatial modeling and simulation platform for community resilience planning
- Completed the RIoT Accelerator Program
- Completed NSF Innovation Corps (I-Corps), conducting 35 customer discovery interviews

Ph.D. Candidate and Research Assistant Aug 2018 to May 2023

Center for Geospatial Analytics, North Carolina State University

- Created OpenPlains, an open-source platform for geospatial participatory modeling built on GRASS
- Developed Rapid-DEM, a prioritization tool for UAS-driven DSM updates through satellite change detection and data fusion (Google Earth Engine, GRASS, Google Cloud)
- Coupled spatially explicit surveys with fuzzy cognitive maps to study how spatial scale affects mental models of stormwater management

- Performed hydrological modeling of surface runoff and flooding using both process-based and geometry-based simulations (GRASS, Python, R)
- Extracted geospatial features for a Bayesian spatial-temporal nutrient loading model for water-supply watersheds
- Teaching Assistant: Grand Challenges in Geospatial Analytics; Geospatial Computation and Simulation; Geospatial Modeling and Analysis

Earlier Professional Experience

Consultant, Technical Lead, and Software Engineer

Oct 2015 to Jan 2020

JouleBug, Raleigh, NC

- Technical lead across iOS, Android, web, and microservice stacks; architected AWS infrastructure; led API development in Scala (Play Framework) and Node.js

GIS Programmer, Analyst, and Specialist

Jan 2014 to Sep 2016

City of Raleigh, Public Utilities Department, Raleigh, NC

- Technical lead for enterprise distribution-network GIS applications (ArcGIS, Oracle Spatial, ArcGIS Server); released the open-source Angular-ArcGIS-Server library

GIS Technician and Intern

Jun 2010 to Aug 2013

City of Asheville, Asheville, NC

- Mapped approximately 15,000 interments at historic Riverside Cemetery using paper records, field surveys, orthoimagery, and ground-penetrating radar
- Developed 20 cm orthoimagery from helium-balloon aerial photography, improving prior 6-inch resolution
- Digitized capital infrastructure and established GTFS feed for Asheville bus routes on Google Transit and OpenTripPlanner

BOOK CHAPTERS

1. White, C. T., Mitsova, H., Neteler, M., Petrasova, A., & Hofierka, J. (2026). Chapter 12: Geomorphometry in GRASS. In H. I. Reuter, C. H. Grohmann, & V. Lecours (Eds.), *Geomorphometry: Concepts, Software, Applications* (2nd ed., Vol. 37, forthcoming September 2026). Elsevier. Paperback ISBN 9780443333767; eBook ISBN 9780443333774.

PEER-REVIEWED JOURNAL ARTICLES

1. White, C. T., Petrasova, A., Petras, V., Tateosian, L. G., Mitsova, H., & Meentemeyer, R. K. (2023). An open-source platform for geospatial participatory modeling in the cloud. *Environmental Modelling & Software*, 167, 105767. <https://doi.org/10.1016/j.envsoft.2023.105767>
2. White, C. T., Reckling, W., Petrasova, A., Meentemeyer, R. K., & Mitsova, H. (2022). Rapid-DEM: Rapid topographic updates through satellite change detection and UAS data fusion. *Remote Sensing*, 14(7), 1718. <https://doi.org/10.3390/rs14071718>
3. White, C. T., Mitsova, H., BenDor, T. K., Foy, K., Pala, O., Vukomanovic, J., & Meentemeyer, R. K. (2021). Spatially explicit fuzzy cognitive mapping for participatory modeling of stormwater management. *Land*, 10(11), 1114. <https://doi.org/10.3390/land10111114>

In Preparation

1. White, C. T., Regmi, P., Reckling, W., & Mitsova, H. (in preparation). Topographic change assessment for hazard mitigation planning for Hurricane Helene.
2. Regmi, P., White, C. T., & Mitsova, H. (in preparation). Geomorphic impacts after Hurricane Helene: A multi-sensor geospatial analysis in western North Carolina, United States. Target journal: *Natural Hazards*.

PEER-REVIEWED CONFERENCE PROCEEDINGS

1. White, C. T., & Haedrich, C. (2025). Workshop: Propagating DEM uncertainty to stream extraction using GRASS. *Geomorphometry 2025*, Perugia, Italy, June 9 to 13, 2025. <https://doi.org/10.5281/zenodo.15283714>
2. White, C. T., Petrasova, A., Reckling, W., & Mitsova, H. (2020). Automated land cover change detection through rapid UAS updates of digital surface models. *ISPRS Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLII-3-W11, 155 to 159. <https://doi.org/10.5194/isprs-archives-XLII-3-W11-155-2020>

TUTORIALS AND EDUCATIONAL PUBLICATIONS

1. White, C. T. (2026). *Geomorphometry with GRASS*. Zenodo. <https://doi.org/10.5281/zenodo.19403682> (companion record: <https://doi.org/10.5281/zenodo.19403683>)
2. White, C. T. (2025). *GRASS Addon Development with Python*. Zenodo. <https://doi.org/10.5281/zenodo.17782272> (companion record: <https://doi.org/10.5281/zenodo.17782271>)
3. White, C. T. (2024). *Geoprocessing with GRASS GIS and Jupyter Notebooks*. Zenodo. <https://doi.org/10.5281/zenodo.13367159>
4. White, C. T. (2024). *Intro to Geoprocessing Workshop*. Zenodo. <https://doi.org/10.5281/zenodo.10764925>

DOCTORAL DISSERTATION

White, C. T. (2023). *Facilitating civic engagement in rapidly urbanizing regions through geospatial participatory modeling* [Doctoral dissertation, North Carolina State University]. ProQuest Dissertations and Theses. ISBN 9798379880231.

SOFTWARE AND DATA RELEASES

GRASS project contributions (OSGeo)

- **Core (OSGeo/grass):** Code contributions to GRASS core.
- **Documentation:** Led the redesign of GRASS documentation and authored substantial portions of the new Markdown-based documentation (NSF POSE, 2024 to 2025).
- **Website (OSGeo/grass-website):** Led the redesign of the GRASS project website and deployed the refreshed GRASS branding (new logo, colors, and language) across the website, tutorials, and documentation following PSC approval in 2025.
- **Addons (OSGeo/grass-addons):** Author of `r.random.walk`, `r.hand`, `v.surf.rst.cv`, `t.stac`, and `r.in.ssurgo` (imports USDA NRCS SSURGO soil data; associated with USDA NRCS Award #NR233A750023C043).
- **Tutorials (OSGeo/grass-tutorials):** Designed and set up the Quarto-based GRASS tutorials website.
- **Addon cookiecutter (OSGeo/grass-addon-cookiecutter):** Contributor.

Archived software releases with DOI

- White, C. T. (2023). *tomorrownow/TomorrowNowApp: v0.0.1 (alpha.0)*. Zenodo. <https://doi.org/10.5281/zenodo.7702537>
- White, C. T. (2023). *tomorrownow/django-actinia: v0.0.1 (alpha.0)*. Zenodo. <https://doi.org/10.5281/zenodo.7702948>
- White, C. T. (2023). *tomorrownow/grass-stac-extension: v1.0.0*. Zenodo. <https://doi.org/10.5281/zenodo.7703038>

Other open-source projects

- **OpenPlains (OpenPlainsInc/OpenPlains):** open-source geospatial participatory modeling platform
- **learning.openplains.com:** technical tutorial site covering open-source geospatial workflows, including NISAR terrain change detection in Quarto, CONUS-scale binary raster mask construction, and GDAL and GRASS compression techniques
- **Angular-ArcGIS-Server (2015):** open-source AngularJS integration library for ArcGIS Server

GRANTS AND FUNDED RESEARCH

- **NSF POSE Phase II:** Growing GRASS OSE for Worldwide Access to Multidisciplinary Geospatial Analytics. NSF Award #2303651. NC State University (lead), with subawards to Arizona State University, New Mexico State University, and Yale University. PI: Helena Mitsova. Co-PIs: Huidae Cho, C. Michael Barton, Vaclav Petras, Anna Petrasova. Total awarded: \$1,566,354 to NC State (initial \$1,499,987 plus FY24 supplement \$66,367). Period: September 2023 to August 2026. Role: Senior Personnel.
- **USDA NRCS:** Adaptation of the path sampling water flow simulation model (SIMWE) for the integration of Dynamic Soil Survey data. USDA NRCS Award #NR233A750023C043. NC State University. Role: Lead researcher.
- **Cesium Ecosystem Grant (in application):** Interactive 3D and 3D Tiles capabilities for `grass.jupyter`. NC State University, up to \$50,000.

- **NSF R212:** Co-Developing Mountains-to-Coast Resilience Solutions through Integrated Science and Practice. NSF Award #2519237. Regional Resilience Innovation Incubator (Phase 1). NC State University. Role: Senior Personnel.

INVITED TALKS

1. White, C. T. (2026, April 1). *Geomorphometry with GRASS*. Coffee Talk, International Society for Geomorphometry.

CONTRIBUTED TALKS, POSTERS, AND ABSTRACTS

1. Regmi, P., White, C. T., & Mitsova, H. (2026, April 26 to 30). *Assessment of landscape disturbance after Hurricane Helene using multi-sensor elevation and imagery data*. IALE-North America Annual Meeting, Athens, GA. Oral presentation.
2. Regmi, P., White, C. T., & Mitsova, H. (2025, December). *Geomorphic impacts and terrain recovery after Hurricane Helene: A multi-sensor geospatial analysis in western North Carolina*. American Geophysical Union Fall Meeting 2025. Poster.
3. White, C. T., & Haedrich, C. (2025, September). *State of GRASS*. FOSS4G North America 2025.
4. White, C. T. (2025, April 16). *Topographic impacts of Hurricane Helene in western North Carolina*. IALE-North America Annual Meeting. <https://doi.org/10.5281/zenodo.15237651>
5. White, C. T. (2024, December). *Sensitivity analysis of spatial scale and particle density on overland flow pattern accuracy and computational demand*. American Geophysical Union Fall Meeting 2024.
6. White, C. T. (2024). *GRASS GIS as a geospatial processing engine*. FOSS4G 2024.
7. White, C. T. (2023, December 12). *A web-based platform to democratize geospatial participatory modeling in the cloud*. American Geophysical Union Fall Meeting 2023. <https://doi.org/10.5281/zenodo.10456565>
8. White, C. T. (2022, November 14). *Assessing the efficacy of BMPs on controlling nutrient loading through Bayesian watershed modeling*. Water Resources Research Institute Annual Conference.
9. White, C. T. (2022). *OpenPlains: is this the new Web GRASS?* FOSS4G Firenze 2022.
10. White, C. T., Petrasova, A., Reckling, W., & Mitsova, H. (2019). *Automated land cover change detection through rapid UAS updates of digital surface models*. Pecora 21 / ISRSE 38, Baltimore, MD.
11. NSF Smart and Connected Communities PI Meeting (Smart Cities Connect), Denver, CO (2019).
12. NC GIS Conference 2019.
13. White, C. T. (2015). *Angular-ArcGIS-Server*. NCAUG Conference, Carolina Beach, NC.

WORKSHOPS AND TUTORIALS DELIVERED

1. White, C. T., & Haedrich, C. (2025, September). *GRASS addon development*. Workshop, FOSS4G North America 2025.
2. White, C. T., & Petras, V. (2025, March 18). *Geospatial Data Science with GRASS*. Pre-conference workshop, NCGIS 2025 (4 hours).
3. Contributor to *Beyond the GIS Map: Open Source Tools for Geospatial Research*. Institute for Social Science Research, Arizona State University, March 25, 2025.
4. White, C. T. (2025, February 8). *Introduction to GRASS GIS Geospatial Processing*. SMATHacks hackathon, North Carolina School of Science and Mathematics.
5. White, C. T. (2024, September). *Developing web applications with GRASS GIS*. Workshop, FOSS4G North America 2024, St. Louis, MO.
6. White, C. T. (2024, August 22). *Geoprocessing with GRASS GIS and Jupyter Notebooks*. U.S. Department of the Interior webinar.
7. White, C. T. (2024, March 2). *Introduction to GRASS GIS Geospatial Processing*. SMATHacks hackathon, North Carolina School of Science and Mathematics.
8. Petras, V., & White, C. T. (2023, October). *Develop geospatial workflows and custom tools with GRASS GIS*. Workshop, FOSS4G North America 2023.
9. Briefing: *r.sim.water overland flow visualization with Cesium.js*. USDA Natural Resources Conservation Service.

TEACHING

Instructor of Record, North Carolina State University

- GIS 582, Geospatial Modeling and Analysis (graduate)
- Mapping and Analytics with UAS (graduate)

Teaching Assistant, North Carolina State University (2018 to 2023)

- Grand Challenges in Geospatial Analytics
- Geospatial Computation and Simulation
- Geospatial Modeling and Analysis

Guest lectures

- Graduate seminar on using Claude and Gemini Pro for scientific software development (90-minute lecture, 2025).

MENTORING AND ADVISING

- Pratikshya Regmi, graduate teaching assistant (GIS 582 and GIS 584); research in UAS mapping and topographic change detection; recommended for Geological Society of America travel awards.

PROFESSIONAL SERVICE

- Peer reviewer, Environmental Modelling & Software (2 reviews).

AWARDS, HONORS, AND CERTIFICATIONS

- Certified GIS Professional (GISP)
- Eagle Scout, Boy Scouts of America (2002)

PROFESSIONAL MEMBERSHIPS

- American Geophysical Union (AGU)
- International Association for Landscape Ecology (IALE)
- International Environmental Modelling and Software Society (iEMSs)
- Geological Society of America (GSA)
- International Society for Geomorphometry (ISG)
- Open Source Geospatial Foundation (OSGeo)

SELECTED MEDIA AND OUTREACH

- *Graduation to Vocation: Corey White is Democratizing Geospatial Data*. College of Natural Resources News, NC State University, May 2023.
- learning.openplains.com: technical tutorials on open-source geospatial workflows, including NISAR terrain change detection, CONUS binary raster mask construction, and GDAL and GRASS compression techniques.

TECHNICAL SKILLS

Geospatial and scientific computing: GRASS (core and addon development), GDAL, PROJ, GEOS, LiDAR and STAC, raster and vector analysis, hydrological and erosion modeling, DEM and DSM processing, UAS photogrammetry, geomorphometry, topographic change detection including Nuth and Kaab co-registration, Google Earth Engine.

Programming: Python (NumPy, SciPy, Jupyter, Quarto, Django, FastAPI), R (RStudio, gstat), Rust, C, CMake, JavaScript and TypeScript (React, Vue, AngularJS, Node.js), Cesium.js, Three.js, Scala (Play Framework), Java, Swift, Objective-C.

Databases: PostgreSQL with PostGIS, Oracle Spatial, MongoDB, SQLite, Google BigQuery.

Cloud, DevOps, and infrastructure: Docker, Terraform, GitHub Actions, Jenkins, Amazon Web Services, Google Cloud Platform, CI/CD.

AI-assisted development and LLM tooling: Claude Code, Claude (Anthropic API and Model Context Protocol), Gemini Pro, LLM-assisted scientific software engineering.